

A FIELD MANUAL

Zombieproof Water

How to Keep a Cup When the Tap Goes Quiet

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The dead may walk. The water still needs a boil.

— Written on the barrel, in chalk, then properly

A FIELD MANUAL IN SEVENTEEN CHAPTERS

Zombieproof Water

How to Keep a Cup When the Tap Goes Quiet

AUTHOR

PUBLISHER

CITY

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This is a work of humor and amateur curiosity. It is not a drinking-water licence, medical advice, a well-driller's certificate, a lightning code, or permission to plumb a boiler. Public-health guidance changes. The label on the bottle and the instruction from your local authority win every argument with a number in this book. Flood water, chemical water, and fire-sprinkler water are not a dare. The undead are a metaphor. The germs are not.

Illustrations made for this edition. Author, dedication, and publisher left blank to be filled in ink.

PRINTED IN

Dedication

For

NAME

AND, IF YOU LIKE, A LINE

AND ONE MORE

Early word from the tap

Stand-in notices, so the page has a shape. Cross them out. Put real readers here.

I read the first page aloud to a rain barrel. It did not become sterile, which I am choosing to take as honesty.

— Gutter, name withheld, cup still empty

Finally a water book that admits the horde is not the hard part. The hard part is the roof's first bad mood, which you do not drink.

— A person who would like their cistern withheld

I fell asleep in the slow-sand chapter and woke convinced I had a German roommate. The roommate is the biology. I kept the lid on.

— The Committee for Slightly Steadier Schmutzdecke

The pictures understand the assignment. The undead are holding the cups upside down. I kept it between the bleach and the good kettle.

— Go-bag inspection, label still readable

Heard a bit of the sprinkler chapter read in a yard. Somebody laughed once, which is the correct number of times, and then did not drink from the riser.

— Workshop, identity unconfirmed, thirst intact

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Foreword

Someone always asks, at the bad moment, whether it is safe. They mean the tap. They mean the barrel. They mean the puddle the street has offered, which is not an offer.

This manual is an answer with a lid on it. It begins where water actually begins after a failure, which is not with a clever filter but with a decision not to be poisoned by the thing you were trying not to die of thirst over. Then it says where water comes from when the mains have gone quiet: a roof, a foggy hill, a pipe that has been stagnant for years and should be treated as a rumour, and the weather, which lies.

You will find boiling told slowly, because it is still the method that does not require you to guess the strength of a bottle. You will find bleach, because sometimes the kettle is a luxury, and the bottle must be the one you can name. You will find slow sand, pocket filters, and storage, each with the thing it does not do written in the same size as the thing it does. Later chapters wander into gravity, clay pots for gardens, and hot water, and then stop, on purpose, before they become a plumbing licence.

The zombies are a convenience. They stand in for hurry. A boil-water notice nobody read, a roof wash drunk because it looked clear, a person who wants the barrel to be a personality instead of a container. If you are reading this while the world is, against the odds, still selling unscented bleach, the zombies will not mind. The germs are the same on a Tuesday.

A word about the pages that mention doses, wells, and heat. Curiosity is not a licence. Do not drink fire-sprinkler water because a chapter explained why it is filthy. Do not build a woodstove coil because a chapter explained why the tank has to sit above the fire. Do not pour a sack of pool chemicals into a well because the grid is having a mood. The jokes are jokes. A water

main can fail and the rules about infants, pregnancy, and chemical contamination can still be standing. Cryptosporidium does not care that you labelled the jug.

What I hope you keep is not a single gadget. It is the habit of a cup you understand: where it came from, what you did to the living things in it, what you could not do to the metals and the solvents, and how you stored it so tomorrow's cup is not a fresh argument. Water is poor company and an excellent servant. It does not replace the person who treated it. It only makes it possible for the morning to include a drink.

Leave your name off the title page until you want it there. The kettle does not require a byline. It requires a minute, a little discipline, and somebody who did not skip the page.

The foreword writer signs here.

NAME

PLACE AND DATE

A note on the drafts

This manual was expanded from a pile: emergency disinfection cards, a rainwater manual, notes on slow sand, a fog-mesh paper, disaster pages about floods and fire and the tap afterwards, and a foundation draft that already knew the unglamorous order of operations. The draft was careful in the way a briefing is careful. What it lacked was the yard. The jokes, the zombies, and the habit of saying a hard rule twice in a human voice were put in afterwards. The public-health spine was not loosened to make room for them.

Where the notes gave a number — a minute of boil, eight drops, a runoff coefficient, a foot of head — the number is still here, and it is still labelled as indicative. Household bleach is not one chemical wearing one hat. Modern bottles wander from about five percent to about nine, and some are not chlorine at all. If the label on your bottle and the table in this book disagree, the label wins, and your local health department wins over both. A historical starting point is not a recipe.

The chapters at the back are refusals as much as they are lessons. The thermosiphon is described so you will recognise one, not so you will plumb one. The well is described so you will wait for a test, not so you will shock-chlorinate a borehole from a humor book. The lightning note is here because a metal roof and a full tank are a serious electrical neighbour, and the neighbour has a code and a profession.

You can read it straight through, or raid the contents the way you would raid a cupboard. The first part is how not to die of enthusiasm. The middle is where water comes from and how you make the biological part of it lonely. The last part is how to keep a cup, how to move it without a bill from the grid, and what this edition will not pretend to teach.

There is a dull warning, and it is dull on purpose. Do not drink flood water to see if you are hardy. Do not use scented bleach because it was nearest the sink. Do not treat a humor book as a boil-water notice. The undead are a metaphor. A cistern of asphalt-roof wash is not.

The dedication, the author's name, and the publisher are blank on purpose. Write them in when you know them. Ink, not pencil. Pencil is for first-flush marks.

PART I

Before the Cup

What water can still kill you with, and how much of it a person actually needs.

CHAPTER 1

The Page You Do Not Skip



Plate 1. The cup stays empty until the water has been introduced.

The dead may walk. A germ does not need their help, and it will not wait while you decide the water looks fine.

Clear is a compliment, not a test. Water can carry bacteria, viruses, and the stubborn cysts of Giardia and Cryptosporidium, and it can carry them while sparkling like an advertisement for honesty. It can also carry things no kettle will evict: fuels, solvents, heavy metals, the antifreeze somebody put in a pipe to survive a winter, the ash and chemistry a fire left on a roof. Boiling, bleach, and a pocket filter are wonderful at some of these jobs and theatrical at others. Everything after this page — the roof arithmetic, the jokes, the zombie holding a cup upside down — assumes you have already decided to remain alive, and to remain un-poisoned, which is a slightly more specific ambition.

Read this once when you are calm. Read it again when you are thirsty, which is when you will be tempted to skip it.

The rules that do not negotiate

If the authorities are still speaking, listen to them. A boil-water notice, a do-not-use notice, and a "bottled water only" notice are not suggestions from a humour book. They are the people who can see the plant. This manual is what you read when the notice cannot reach you, or when you are practising on a Tuesday so the bad day is not your first day.

Use water you trust for drinking, cooking, ice, brushing teeth, and anything a baby will meet. Washing the floor is a different conversation. A baby's bottle is not.

Prefer water that was sealed before the trouble: bottled water, a commercial jug, a cistern you filled and treated while the world was still boring. If you do not have that, treat what you collect. Do not drink flood water. Do not drink water that smells of fuel, solvent, or a chemical plant having a feeling. Boiling, bleaching, and filtering do not make chemical contamination safe. They can concentrate some of it, which is the opposite of a favour.

Never store drinking water in a container that held fuel, pesticide, solvent, or anything you would not lick off a spoon on purpose. Plastic has a memory. So does your liver.

Do not use scented bleach, colour-safe bleach, splashless bleach, or a bottle that does not say it is chlorine. "Bleach" is a marketing family. Sodium hypochlorite is a chemical. If you cannot find the percent of available chlorine on the label, you do not have a dose. You have a vibe.

Do not sprinkle pool shock, calcium hypochlorite granules, or any dry chlorine straight into a cup. It burns. It is mixed, if it is mixed at all, into a stock solution, away from fuel and from enthusiasm, and then the stock is measured. The chapter that says so will say so again. It is still not an invitation.

Do not drink from a fire-sprinkler pipe because it is a pipe with water in it. That water may have sat for years with bacteria, metals, and antifreeze. Bleach does not remove lead, and it does not remove glycol. Chapter 5 is a warning, not a recipe.

Do not plumb a boiler, a wood-stove coil, or a pressure vessel because a later chapter explains why hot water rises. A thermosiphon without a relief valve is a bomb that also does laundry. This book will describe it so you can refuse it.

Do not disinfect a drilled well from a field manual. The people who do that work for a living already find it difficult, which is the EPA's polite way of saying the owner with a funnel is not the hero of that story. After a flood, assume the well is guilty until a test says otherwise.

Infants, pregnant people, and anyone with a thyroid condition should not be handed iodine-treated water as a lifestyle. If you must treat water for a baby and you cannot get a sealed commercial supply, boiling and cooling is the kinder path. Formula instructions and a clinician outrank this paragraph. Iodine is a short, ugly spare, not a pantry.

The procedures in this book are educational. They are copied, in spirit and often in number, from public cards: a minute of rolling boil, three minutes above the high country, unscented household bleach in the range those cards describe, a wait of half an hour. Labels change. Percents change. If this book and the bottle disagree, the bottle is right and the book is a well-meaning stranger with a damp cuff.

A sentence you can say out loud

Know where it came from. Take the dirt out. Kill what boiling or a named disinfectant can kill. Admit what they cannot. Put a lid on it.

The zombies in these pages are a convenience. They stand in for hurry, folklore, and the part of a person that wants a drink more than they want to be there in the morning. They are not

the hazard. The hazard is the water, the container, and the decision to skip a step because the street sounds busy.

If you remember one line from the whole manual, let it be the one on the barrel, written first in chalk and then properly: the dead may walk. The water still needs a boil.

CHAPTER 2

How Much, and From Where



Plate 2. Three litres, a map, and a zombie drinking from the puddle.

Thirst is a poor planner. It arrives late, speaks loudly, and will try to promote the nearest puddle.

A person at rest in mild weather needs on the order of two and a half to four litres of drinking water a day, more if they are working, more if the day is hot, more if they are ill. That is the cup you argue about. It is not the whole bill. Cooking, washing a wound, washing hands, and not living in your own smell all ask for more, and a garden asks for a shocking amount if you water it like rain that has forgotten how. The survival number and the civilisation number are different. Write both down when you are calm, in litres, not in vibes.

The order of trust, when you still have a choice, is dull and correct.

Sealed water you packed before the trouble. Then water you treated yourself by a method you can explain. Then water you are only using to wash things that are not mouths. Last, and preferably never, water that has been industrial: flood water, sprinkler mains, a sump, a ditch that has met a road.

What "safe" is allowed to mean

Safe, in this book, means safe from the living hazards you can actually do something about: the bacteria, the viruses, and the parasites that boiling, a proper chlorine dose, or a filter rated for them can address. It does not mean safe from lead, fuel, pesticides, or a fire system's antifreeze. Those require you not to use that source, or to use a treatment this book will not pretend you can build on a Saturday. If you cannot say which of those two sentences you are in, you are in the first one. Do not drink it.

Surface water — rivers, ponds, the dramatic puddle — should be assumed to contain faeces, because the world is full of animals and people and both of them are honest about that. Roof water should be assumed to contain the roof: dust, soot, bird lime, and whatever the shingles are made of. Fog water should be assumed to be a small, proud trickle that still needs the same manners as a roof. Tap water, after a flood, a hurricane, a wildfire near the intake, or a plant that has lost power, should be assumed to be a rumour until someone whose job it is says otherwise.

A small map

When you look at a place, look for water in this order, and write it down so the thirsty version of you cannot innovate.

The bottles and jugs already in the house. The hot-water cylinder, if you can drain it cleanly and you are sure it was potable when it filled — it is a tank, not a miracle, and it may be rusty or hot enough to scald. The rain barrel, if you built it for drinking

and you are willing to treat it. The roof you could collect from, if you have a clean container and a way to throw away the first wash. A foggy ridge, if you live on one and you have mesh, and you have made peace with modest yields. A neighbour's treated store, if the neighbour is still a neighbour.

Not on the map: the flood in the street, the sprinkler riser, the chemical store, the ornamental pond that has been a bird's entire civilisation, the water from a radiator.

A zombie will drink from the puddle. That is how you know it is a zombie, or in a hurry, which for the purposes of this chapter is the same profession.

The rest of the book is the methods. This chapter is only the decision to have a method before you have a mouthful.

PART II

Where It Comes From

Roofs, fog, the pipes that have been waiting, and the weather that lies about the tap.

CHAPTER 3

The Roof That Pays Rent



Plate 3. The roof pays rent. The first wash does not come inside.

Rain is the most polite large source you are likely to meet. It arrives already fallen, it does not send a bill, and it has not, usually, spent a decade in a steel pipe making friends with bacteria. It has, however, met your roof, and your roof has a biography.

The yield people actually use is a small piece of arithmetic, and it is indicative, not a prophecy:

gallons, roughly, equal 0.623 times the roof's footprint in square feet, times the rainfall in inches, times a runoff coefficient.

The 0.623 is the dull conversion from a square foot of one inch of rain into U.S. gallons. A thousand square feet and one inch, if every drop behaved, is about six hundred and twenty gallons. It will not. Wind, splash, and the first wash you ought to throw away steal some. Real systems often land around three quarters

to five sixths of the theory. Metric cousins exist; if you think in litres and millimetres, use a local sheet and do not convert twice.

The coefficient is the roof's honesty. A smooth pre-painted metal roof is the good pupil, often around nine tenths, and it sheds less grit. Asphalt shingles are worse company for drinking water: they hold back more, and they can lend the wash heavy metals, petroleum crumbs, and grit. Clay and concrete tile drink a little of the rain themselves and grow moss when the mood takes them. If you can choose a surface for a drinking cistern, choose metal, and still treat the water. "Better roof" is not "sterile roof."

The first wash is not a guest

A dry roof is a plate. Dust, soot, leaves, and the digestive opinions of birds sit on it until the rain edits them into your gutter. That first water is the dirtiest water the storm will offer. Letting it into the cistern is how you grow a biology experiment and then ask a teaspoon of bleach to fight a mud pie.

A leaf screen or a rain head takes the twigs and the tennis balls. A first-flush diverter takes the filthy volume itself. The usual picture is a standpipe that fills with the opening wash until a ball floats up and seats, and the cleaner rain goes on to the tank. A small bleed hole at the bottom empties the pipe over a day or two, so the next storm finds it ready. If the bleed is clogged, you do not have a diverter. You have a statue.

How much to divert depends on how rude the roof is. Open country, little canopy: on the order of one to five gallons per thousand square feet. Trees, birds, a city that sheds soot: more like five to ten gallons per thousand square feet. The research notes also give a pipe's appetite, so you can build the volume on purpose: about a quarter gallon per foot of three-inch pipe, half a gallon per foot of four-inch, and a bit under one and a half gallons per foot of six-inch. These are workshop numbers.

Measure your actual pipe if the civilisation of measuring is still available. Do not drink the diverter. It is doing a job by being discarded.

What the barrel is not

A rain barrel for the garden can be a charming mess. A cistern for mouths wants a lid, a screen on every opening a mosquito could consider a nursery, an opaque wall or a jacket against the light, and food-grade materials. Light grows algae. Algae are not a seasoning. The inlet should not dump onto the outlet, and the outlet should not sit in the sludge at the bottom. A first-flush that you never empty is just a smell with plumbing.

Roof water is soft and useful and still untreated. Bird lime does not become safe because it fell from a height. Chapter 8 and chapter 9 are not optional for drinking. The roof pays the rent. You still boil the tea.

CHAPTER 4

Fog, If the Hill Is Kind



Plate 4. Mesh in the wind. The yield is a polite trickle, not a miracle.

Some hills spend half their lives inside a cloud that has not committed to being rain. The cloud is still water. It is merely shy, and it arrives as droplets small enough to ignore a gutter: on the order of one to forty microns, which is a way of saying you will not catch them in a bucket by staring.

Fog collection is a fence with manners. Mesh stands across the wind. Droplets hit the filaments, join up until they are heavy enough to be a drip, and run down into a trough. The trough is the whole technology's moment of honesty. If the drip falls back into the dirt, you have built a sculpture.

The cloth the experiments like is Raschel mesh, a polyethylene shade cloth. A single layer is the baseline. Two layers, or a puffier three-dimensional knit, give the droplet more places to change its mind and become liquid. In one set of trials a multi-layer or three-dimensional arrangement gathered more than

twice the water of a single sheet — on the order of ninety-nine millilitres against thirty-nine, in that test's little world. That is a ratio, not a promise that your hillside will fill a bath. Yields depend on how often the fog actually visits, how wet it is, and whether you pointed the mesh at the wind or at the view.

Point it at the weather, not the sunset

Face the fence across the prevailing damp wind, not along it. Put it where the cloud meets the ground often, which is usually higher and ruder than the picnic spot. Keep the trough clean. A mesh that sags into a pocket grows algae and a smell, and then you have invented a swamp with ambition.

Fog water is not distilled holy water. It has met air, and air has met birds, dust, and whatever the upwind valley is burning. Treat it before you drink it, the same as roof water. And do not move to a dry ridge because a paper used the number 150 per-cent. A hundred and fifty percent of a trickle is a slightly more confident trickle.

The zombie will stand in the fog with its mouth open. This is not a design. The mesh is the design. The zombie is the control group.

CHAPTER 5

The Pipes That Have Been Waiting



Plate 5. The sprinkler pipe has been waiting. It is not a well.

A town is full of water that is not yours. The most tempting wrong example is the fire sprinkler. It is a pipe, it is full, and in a bad month it looks like a well that learned about ceilings.

It is not a well. Sprinkler lines are closed. The water in them does not circulate. It sits, sometimes for years, at room temperature, which is the temperature bacteria write home about. Anaerobic films grow. Iron-eating and sulphate-reducing organisms make slime and pinholes. Legionella and ordinary coliforms are not a surprise in that kind of stillness. Metals leach: iron, and more importantly lead, copper, and zinc, at levels the drinking-water limits were invented to forbid. In cold countries the pipe may also contain antifreeze — propylene glycol, ethyl-

ene glycol, or a corrosion inhibitor with a marketing department. High glycol is toxic. It is also, in a fire, not the thing you want to be casually decanting near a flame.

Bleach will not fix this. Chlorine is a weapon against many living things. It is not a magnet for dissolved lead, and it does not cancel a solvent. The notes in the pile are blunt: if you were going to try to make such water drinkable you would be talking about settling, activated carbon, and distillation, and even then some chemicals travel with steam or refuse to be a weekend. That is an industrial sentence. It is not a shopping list. This book will not walk you through a still. A still that is also a pressure vessel is how garages leave the street.

Backflow is a door, and it should stay a door

There is a second lesson hiding in the sprinkler, and it is about the ordinary house. A sprinkler, a hose, a boiler, and a drinking tap should not be able to kiss. Backflow preventers exist so that a stagnant, chemically peculiar pipe cannot wander backward into the water you actually drink. If you are still in a world with inspectors, do not defeat those valves because a hose would be more convenient. If you are not, do not invent a shortcut that ties a fire main to a kitchen tap "just while we sort it." The shortcut is the contamination.

If the only water you can see is in a sprinkler riser, you are in a worse day than this chapter can repair. Look again for the roof, the bottled store, the neighbour, the kettle you can still boil a ditch in — a ditch is a miserable source, but it is a biological miserable, and biological miserable is a problem boiling understands. Glycol is not.

Leave the riser alone. Let the zombie try it. That is what the pictures are for.

CHAPTER 6

After the Weather



Plate 6. After the weather, the tap is a rumour until someone official says otherwise.

Weather does not only take the roof. It takes the meaning of the tap.

After a flood, the water in the street has been a mixture of sewage, soil, fuel, and whatever the upstream shed was storing. It is not a source. It is a reason to wash your hands with water you already trust, and to stay out of it if you can. Homes that have been under water are a different hazard again: electricity that may still be lively, mould, and the quiet damage to a private septic tank or a well. A flooded well should be assumed contaminated. Shock-chlorinating a drilled, driven, or bored well is a contractor's job in the EPA's own pamphlet, because the owner usually cannot get the disinfectant everywhere it has to go. Do not drink it until someone tests it. Do not "just boil a little" from a well that has also met a fuel tank. Boiling is not a spell against petrol.

Hurricanes and tropical storms knock plants offline and push seawater and debris into intakes. The honest posture is the one the health cards repeat until it sounds like a proverb: use sealed bottled water if you have it; otherwise boil, or disinfect with a method you can name; and obey the local notice if a local notice exists. A tap that runs clear after a storm has not been interviewed. Clarity is still a compliment.

Wildfire is sneakier. Ash, fire retardant, and the burned contents of a shed can land on roofs and in open tanks. A cistern without a lid is a bowl. If the fire came through, assume the roof wash is chemical until you have a reason not to, and do not let the first rains after a burn go straight into the drinking tank. Empty the diverter. If the tank itself was open to the ash, do not try to bleach the ash into politeness.

Earthquakes break pipes and mix the potable main with the rest of the ground. Tsunamis are salt, sewage, and ruin; the sea is not a cistern you can filter into a cup. Tornadoes throw the container across the county and leave you with whatever sealed bottles survived the cupboard. In every one of these, the pattern is the same. The infrastructure's promise has paused. Your nose is not a laboratory. Fuel-smell, chemical-smell, and "it came out of the flood" are all stops, not puzzles.

Septic tanks do not become charming

If dirty water has washed over a septic tank, a soakaway, or the lid of either, keep people and animals off it, and do not flush your way back to normal. A system that has been under water can push sewage the wrong direction. That is a plumber and a public-health problem. It is also a reason not to drink from a shallow well downhill of it, no matter how rustic and honest the well looks in the evening light.

The zombie wades. You do not. When the weather has had an opinion, the cup you want is the one you sealed before the opinion started.

PART III

Making It Drinkable

Settle it, boil it, name the bleach, and do not trust a pocket filter to be a priest.

CHAPTER 7

Clear It Before You Trust It



Plate 7. Settle it. The cloth is a manners lesson, not a blessing.

Disinfection is a conversation with living things. Dirt interrupts.

Cloudy water hides microbes inside particles and eats chlorine before the chlorine can do its job. The public cards all say the same dull thing first: if it is cloudy, let it settle, or pour it through a clean cloth, a coffee filter, or a proper prefilter, and draw off the clearer water for the boil or the dose. Settling is not purification. It is manners. An hour in a clean bucket, the sludge left behind, the clearer top poured off gently, is already a different liquid from the one you dipped out of a ditch in a panic.

A cloth has holes the size of gossip. It will stop twigs, some grit, and the more theatrical insects. It will not stop bacteria, viruses, or cysts. If you tell yourself the T-shirt was the treat-

ment, you have joined the zombie, who is currently wringing the same T-shirt into its mouth.

What you are allowed to call "clear"

Clear enough to treat means you can see through it in a glass, more or less, and it is not the colour of tea made from a road. It does not mean potable. After you settle, you still boil or you still dose, unless the water was already from a source you trust. Colour from leaves is not the same as colour from a factory, and your nose gets a vote: fuel, solvent, a sharp chemical edge, and you stop. No cloth fixes that. You find another source.

Keep the settling bucket food-grade and boring. Rinse the cloth with treated water if you have any, and do not use the cloth that wiped the sump. Label the "dirty" bucket and the "clear" bucket so a helpful person cannot merge them in an act of hospitality.

If you have a proper sediment filter, of the sort sold to go in front of a drinking system, use it as the manufacturer says, and replace it when it clogs rather than when it becomes a pet. A clogged filter is a bypass waiting for an impatient person.

The schmutzdecke and the pocket hollow-fibre come later. They are finer tools. This chapter is only the refusal to dose a mud puddle and call it chemistry.

Let it sit. Pour off the top. Then, and only then, introduce it to the kettle.

CHAPTER 8

The Kettle Is the Grown-up



Plate 8. One minute. Three if the hill is high. The kettle does not negotiate.

If you can boil, boil. It is the method with the fewest opinions about brands.

A rolling boil — big bubbles, the real thing, not a shy shimmer — for one minute is the figure on the emergency cards for killing bacteria, viruses, and parasites, including the cysts that chlorine sulks at. Above about six thousand five hundred feet, where the water boils cooler because the air is thinner, those same cards say three minutes. The research notes sometimes draw the line nearer five thousand feet or a thousand metres. Mountains are not a place to economise. If you are high enough to wonder, boil for three.

Let it cool, with a lid, in a clean container. Do not add ice made from water you have not met. Do not stick a finger in "to test it," unless the finger has a retirement plan.

Boiled water tastes flat because you drove the dissolved air out of it. Pour it back and forth between two clean vessels, or leave it an hour, or add a pinch of salt to a litre if you want it to taste like water instead of like a lecture. That is cosmetics. It is not a second treatment.

What the kettle refuses

Boiling does not remove metals, salts, fuels, or most chemical pollutants. It can make them a little stronger, because some of the water leaves as steam and the contaminant stays. If the source was a ditch of manure, the kettle is your friend. If the source was a ditch of diesel, the kettle is a concentrator. Walk away.

Boiling also does not protect the water after it cools. A clean cup can be re-dirtied by a dirty ladle, a thirsty pet, or a lid left off in a room of flies. The boil is a moment. Storage, in the next part of the book, is the rest of the week.

For a baby, if sealed water is gone, boiled and cooled is the path this book will stand next to. Do not boil it down to a syrup. Do not add bleach to formula "to be sure." Follow the formula tin and a clinician if you can get either. Cooling matters: a scald is not a purification step.

Fuel for the boil is part of the method. A kettle on a stove indoors is ordinary life. A charcoal burner in a closed room is the engine book's carbon monoxide, wearing a saucepan. Do the burning where the exhaust can leave. The zombie will boil a pan in a cupboard. You will not.

One minute. Three if the hill is high. Lid on. Then it is water you can explain.

CHAPTER 9

Bleach You Can Name



Plate 9. Unscented, named, and left alone for half an hour.

Sometimes the fuel is gone and the germs are not. Then you want a disinfectant you can point at, which means unscented household chlorine bleach, not the lavender one, not the "splashless" one, not a bottle that boasts of being gentle.

The card in the source pile, the one meant to be taped inside a cupboard, says this for a bottle in the five-to-nine percent range: eight drops, or a little less than an eighth of a teaspoon, in one gallon of clear water. If the water is cloudy, sixteen drops, or a quarter teaspoon — but you should have settled it first, so that you are not spending the dose on mud. Stir. Wait thirty minutes. It should smell faintly of chlorine. If it does not smell at all, the dose was eaten by dirt or by an old, tired bottle; repeat it and wait again. If it smells like a swimming pool's angry cousin, pour it between clean jugs and let it stand open to air until it is only a hint.

An older table, still in the pile, splits the dose by percent, and it is worth knowing because bottles wander:

Around four to six percent: about two drops per quart, or eight per gallon, which is the same neighbourhood as the card. Around one percent: much more, on the order of ten drops per litre. Around ten percent: less, on the order of one drop per quart. If you truly cannot read a percent, the old note says ten drops per quart or litre of water you have already filtered and settled, then the smell test. Doubling for cold, cloudy, or coloured water shows up in both traditions. Cold means roughly below ten degrees, when the chemistry dawdles.

These numbers are indicative. The label on the bottle and the health department currently in charge of your county outrank a book. Some modern "concentrated" bleaches sit near eight percent; the five-to-nine card was written to cover that band with one instruction. If your label prints its own water recipe, use the label.

What chlorine will not do

Chlorine, given time, is excellent against a great many bacteria and viruses. It is a poor oath against Cryptosporidium. That parasite is why the kettle still outranks the bottle when you can light a fire, and why a filter that is actually rated for cysts still matters if chlorine is your only tool and the source is a ditch. Chlorine also does nothing useful about lead, fuel, or pesticides. A faint chlorine smell means you have residual disinfectant, not that you have removed a solvent.

Bleach itself gets old. Heat and light spend it. A bottle that has sat in a shed for a year may be a ghost of the percent on the label. Keep it cool, dark, and capped, and do not trust a half-remembered jug the way you trust a new one.

Dry calcium hypochlorite — pool shock, high-test hypochlorite, the granules that last for years if they stay dry and cool — is a stockpile chemical, not a pinch. The old recipe makes a stock:

about one heaping teaspoon, around seven grams, stirred into two gallons of clean water, giving a solution near five hundred milligrams per litre. You then add that stock at about one part to a hundred parts of the water you are treating: a pint of stock in about twelve and a half gallons, or the same ratio in whatever vessels you actually own. Mix, wait thirty minutes, look for the faint smell. Never toss the granules into the drinking water neat. Never store the granules next to fuel, oil, or a cloth that wants to become a fire. It is an oxidiser. It has hobbies.

Iodine, if it is all you have, is the spare the cards allow and the one this book likes least. Five drops of two-percent tincture per quart or litre of clear water, ten if cloudy, thirty minutes, is the figure in the notes. Tablets follow their packet and not your memory. Do not make iodine a daily habit, and do not hand it to a pregnancy or a thyroid as if it were salt. Taste lingers. So can trouble.

Chlorine dioxide tablets, if you bought them for this exact job, are used as the packet says. The packet is the dose. A blog is not.

Thirty minutes is part of the chemistry. Drinking it at minute two is how you meet the germs socially before the introduction is finished.

CHAPTER 10

Slow Sand and the Schmutzdecke



Plate 10. The schmutzdecke is a roommate. It is not a certificate.

There is a filter that is mostly sand, gravity, and a living skin, and it has been making water kinder for a long time. The grand version is a slow sand bed at a treatment works. The household cousin people mean, when they are off the grid, is a biosand filter: a box with gravel at the bottom, coarser sand in the middle, fine sand on top, and a flow so slow it feels like a philosophy.

Four things happen at once, if the filter has been built to a real design and has been allowed to grow up. The sand strains out particles and many of the larger cysts. Grains, and the iron films on them, catch microbes by stickiness as much as by size. In the top centimetre or two a biological carpet forms — algae, bacteria, protozoa — which the trade calls the schmutzdecke, a word that sounds like a roommate because it is one. That layer eats

and competes. Deeper down, in poorer oxygen, a lot of what was trapped simply fails to thrive. A mature, slow, properly minded filter can take out the great majority of protozoa and a large share of bacteria and viruses. The figures in the notes — better than 99.9 percent of the protozoa, and often 90 to 99 percent of the bacteria and viruses — belong to filters that are actually working, not to a bucket of beach sand with a tap.

It is not a blessing and it is not finished

A new biosand filter is a pile of geology. The schmutzdecke takes time, on the order of weeks, and it wants to stay wet. If you let the sand dry, you kill the roommate and you are back to straining. If you pour water through as fast as a tap, you are not doing slow sand. You are rinsing. Designs from the organisations that publish them — bed depths, grain sizes, a standing head of water, a flow you can count in litres per hour and not in excitement — are the designs. This chapter will not invent a new one in a humorous voice. A humorous sand filter is how people get confident and sick.

Even a good one is a barrier, not the whole fence. The public-health habit is multi-barrier: settle, then a real filter, then boil or a chlorine dose you can name, especially for viruses, which are small and rude. Sand does not remove dissolved metals or fuel. If the water went in smelling of a solvent, it will come out with a cleaner complexion and the same bad character.

Maintenance is mostly restraint. You do not dig the sand out because it looks mucky on top; that muck is the job. When the flow slows too far, the published designs tell you how to stir and discard only the very top. A filter with a cracked body, a dried bed, or a bypass hose "for when we are in a hurry" is a bypass.

Name the roommate if you want. Do not ask it for a certificate. After the sand, the kettle still gets an introduction.

CHAPTER 11

The Filter in a Pocket



Plate 11. The pocket filter stops the large gossip. The viruses walk through.

A hollow-fibre filter you can put in a pack is a wonderful thing against the monsters it was built for, and a door left open against the ones it was not.

The fibres people sell for camping often have pores around a tenth to two tenths of a micron. That is small enough to stop protozoa and a great many bacteria, if the filter is intact, if you follow the maker's backflush and lifespan, and if you do not punch a hole in it with a heroic squeeze. Viruses are smaller still, often a few hundredths of a micron. They walk through a fibre that was never asked to stop them. The emergency card is plain about the family of gadgets: most portable filters do not make water safe from viruses by themselves, and even the ones that remove parasites may need a second step. Read the label. "Removes Giardia and Crypto" is a specific boast. "Water filter" is a mood.

So the pocket filter is a clarifier and a parasite gate. The boil, or a chlorine or chlorine-dioxide step done by the label, is what you add when the source might carry viruses, which is any source that has met people or animals. Chemical contamination, again, walks through. A filter stuffed with a little activated carbon will improve taste and can grab some organic chemicals for a while, until the carbon is full, at which point it becomes a small black pebble. Carbon is not a residual. It does not keep working in the bottle overnight, and it does not make a fuel spill into tea.

Care and feeding

Do not let a hollow-fibre freeze if the maker says freezing bursts it. A burst fibre is a short circuit. Do not filter hot water unless the device says it can. Do not assume the squeeze bag's last user had a philosophy of hygiene. Backflush on the schedule, with the cleanest water you have, so you are not packing the dirt deeper and calling it maintenance.

UV pens exist, in the travel kits, and they have their own demands: clear water, the right dose of light, batteries that are not a rumour, and no shady corner of the bottle the light missed. They leave no residual. A dirty finger afterwards undoes them. If you own one, the manual that came with it outranks this sentence.

The zombie taps the filter against a rock "to test it." You will not hear a difference when a fibre snaps. Treat the device as a tool with a life, and treat its output as water that has had one barrier, not as a sacrament.

Pocket first if the water is wild and full of cysts. Kettle or a named disinfectant after, if you cannot swear the source was only pretty. Then the lid.

PART IV

Keeping It

A dark cupboard, a little height, and a clay pot that waters roots instead of gossip.

CHAPTER 12

The Dark Cupboard



Plate 12. Opaque, food-grade, and never the can that used to hold fuel.

Treated water has a brief career as safe water, and then the world tries to reapply for the job.

The container is half the method. Food-grade plastic — the bottles and jugs actually sold for water, the heavy polyethylene that is not a leftover from the shed — or stainless steel, or glass. Not a fuel can. Not a pesticide drum rinsed "until it seemed fine." Not a milk jug that has gone soft in the sun if you can help it, and never a jug you cannot close. Opaque, or kept in the dark. Light is how algae get a vote, and some of those blooms make toxins boiling is not a cheerful answer to.

Fill containers when life is boring, date them, and keep them off the concrete floor if the flood is a local hobby. A little air gap at the top is fine; a lid left cocked so the cat can investigate is not. Dip with a clean cup you do not then use to bail the ditch. Better, a tap at the bottom, so nobody's hands go swimming.

The faint smell is a sentry, not a flavour goal

Water you disinfected with chlorine can keep a free residual, and that residual is what stops a casual recontamination from becoming a colony. The notes talk about a free chlorine residual on the order of two tenths to one milligram per litre as the sort of band public practice likes, which in a kitchen without a test kit means a faint smell, not a gasp. If the smell has gone and the water has been sitting, the same notes suggest a small top-up — a drop or two of ordinary household bleach per gallon — then a stir and a wait, and then you taste-check by smell again. If you have a pool test strip that reads free chlorine, this is one of the few times a pool gadget is welcome in the kitchen, used as a reader and not as a dosing philosophy. If you are unsure, boil. Boiling is always allowed to win.

Do not chase a stronger and stronger dose because the water "might as well be safe." Too much chlorine is its own unpleasantness, and it is not how you remove a chemical. Aerate, or start over from a better source.

Clean a container before you trust it for the season: wash, rinse, and a chlorine solution you then rinse out or, if you are storing water that will be drunk with a residual anyway, follow a published storage card rather than inventing a spa. Leave the cap loose nowhere. Label the jug with the date and the method, in ink. The next person, who may be you at 2 a.m., deserves a sentence they can read.

The cupboard is dark. The lid fits. The zombie leaves the cap on the floor, which is why the zombie is not in charge of the water.

CHAPTER 13

Gravity Does Not Send a Bill



Plate 13. A little height, a polite trickle, and nobody climbing the tank.

A tank uphill of a tap is a pump that does not burn fuel. It burns height, and height is honest.

The rule of thumb in the notes: about 2.31 feet of water column give you one pound per square inch, which is the same statement as about 0.433 psi for every foot of height between the water surface and the tap. A barrel on a stand four feet up is under two psi. That is a polite trickle, enough to fill a kettle without dipping, not enough to be a shower. Ten psi, the sort of sad-but-functional spray people mean when they say "low pressure," wants on the order of twenty-three feet of head. Ordinary house pressure, near thirty psi, wants on the order of seventy feet. You will not get seventy feet by standing on a chair. Do not build a tower you cannot climb down from, and do not climb a tank.

Put the tank where a failure soaks the ground and not the spare room. A full thousand-litre tank is a small car's worth of weight pretending to be furniture. The stand is a structure, not a stack of bricks with optimism. Overflow must have somewhere to go that is not your foundation. The outlet sits a little off the bottom so the sludge can have a neighbourhood of its own, and a valve you can actually close is worth more than a second coat of paint.

Pipes are also a height, in disguise

Long, skinny, kinked pipe spends the pressure you bought with the hill. Smooth pipe, not tiny, is the habit: the notes mutter about three-quarter-inch to one-inch runs when the distance gets serious. Every elbow is a little tax. A screen on the inlet, a lid on the tank, and no connection that lets garden water or a sprinkler main sip backward into the store. Gravity will also run the wrong way if you give it a path. Backflow is not a moral; it is a missing check valve.

In freezing weather a raised tank is an ice sculpture with a burst in it. Drain what you must, or bury and insulate on purpose, which is a different design and not a jacket tossed over the lid in November.

Gravity will not treat the water. It will only deliver, faithfully, whatever you put in the tank, including the mistake. Treat first, or treat at the tap with a method you can explain. Then enjoy the fact that the hill is working and the generator is not.

The zombie stacks three crates and calls it pressure. You will count the feet, and you will not stand on the lid.

CHAPTER 14

The Buried Pot



Plate 14. Buried to the neck. The garden drinks. You do not.

A garden drinks like a person who has never heard of a bill. Spray it at noon in a wind and you can lose half or more of the water to air and to the leaves' vanity. The old fix is a pot.

An olla is unglazed clay, the porous kind, buried to its neck among the plants and filled with water, then covered so the neck is not a mosquito's ballroom or a kitten's tragedy. Dry soil pulls water through the clay. Wet soil, after rain, stops pulling. The roots come to the damp and stay there, which means less weed festival on the surface and less evaporation. Gardeners who use them at a sensible spacing talk about cutting a large share of the water they used to throw at the foliage — the enthusiastic notes say up to about seventy percent against wasteful surface watering. That is a garden figure, not a promise about your sand, your wind, or your tomatoes' emotional needs.

The shape you want has a belly and a neck you can find. A glazed pot does not weep; it is a jar, and a jar buried full of water is just a jar you cannot see. A hairline crack empties overnight and then you have a dry pot and a smug weed. Cover the opening. Refill before it goes empty for long, so the clay does not dry and sulk.

This pot is not a cup

Olla water is irrigation water. It is in contact with clay and soil and roots. Do not lift the lid and drink. Do not store your treated drinking water in a buried porous pot and call it a cistern. The cistern is in the dark cupboard, with a lid that means it.

Grey water — the wash water, not the toilet, never the toilet — is a separate argument, and this book will keep it short. If you use wash water on a garden at all, it stays off the leaves you eat, it is not full of salt, bleach, or a mysterious "grease-cutter," and it does not pond against the house. Soil contact and an olla are sometimes how people keep that water away from the salad itself. That is a caution, not a recommendation to water your lunch with the dishes. When in doubt, the olla gets the cleanest water you can spare for plants, and your mouth gets the water from chapter 8.

The zombie buries a glazed vase and waits for the miracle. The miracle was the pores. You checked for the pores before you dug the hole.

PART V

The House After

Hot water you will not invent, jobs that belong to other people, and a morning that still has a cup.

CHAPTER 15

Hot Water You Will Not Invent



Plate 15. The kettle is the hot-water system this book will endorse.

Hot water is a kindness. It washes a wound, it makes a person feel like a person, and it is also how kettles and coils hurt you.

The kettle, heated where the fumes can leave, poured with attention, cooled if it is for a child, is the hot-water system this book endorses. It has a handle. It has a size you can see. It does not store pressure.

There is a more glamorous arrangement called a thermosiphon, and you should know what it is so you can recognise one and refuse to improvise one. Water heated in a coil becomes lighter and rises. Cooler water from a tank above it comes down to take its place. If the tank sits higher than the coil — the notes say the bottom of the tank at least a foot to a foot and a half above the top of the coil — and if every pipe slopes up without an air pocket, the loop can circulate with no pump. If you put the tank

below the fire, the loop stalls and the coil boils. If an air lock sits at a high point, the loop stalls and the coil boils. Boiling in a closed coil is how metal becomes shrapnel.

A real system of this kind, built by someone whose job it is, has a temperature-and-pressure relief valve that can actually open, a discharge pipe that does not point at a face, and usually a mixing valve so the tap cannot come out at scalding temperature. The figure people quote for a relief setting is the sort of thing a manufacturer stamps, not the sort of thing a field guide invents. Scalds happen in seconds. Children and the elderly get them first.

The refusal

This chapter is the end of the instructions. It does not tell you which pipe, which solder, which valve, or which stove. A wood-fired coil is a pressure vessel that happens to be romantic. Pressure vessels sit under codes, relief, and people who can be called back when it weeps. The apocalypse, if it is an apocalypse and not just a long power cut, does not repeal the physics. It only removes the person who would have told you to stop.

If you already own a manufactured, certified solar tank or a stove with a water jacket that a fitter installed, maintain it the way the plate on the side says, and test the relief the way the plate says, and do not cap the relief because it drips. A dripping relief is a sentence. Read it.

Heat a kettle. Have a lid for the cup. Leave the coil to the people who sign for coils.

The zombie plumbs a radiator into a closed drum and calls it a bath. You will hear the bang from the next field, which is too close.

CHAPTER 16

Wells, Lightning, and Other People's Jobs



Plate 16. The well and the lightning are someone else's job. Wave politely.

Some water problems are real and still not yours to solve with a chapter.

A private well is a throat into the ground. In ordinary life it is tested, capped, and kept uphill and far from the septic tank, the fuel store, and the place you wash the tractor. After a flood, an earthquake, or a crack you did not order, it is suspect. Muddy water, a missing cap, a smell, or water that reached the wellhead are all reasons to stop drinking it. The pamphlet in the pile is short on heroics: drilled, driven, and bored wells are best disinfected by a well or pump contractor, because the owner cannot easily wash every surface the water will later touch. Sampling after any disinfection is the point, not the smell of bleach on day one. A boiled kettle from a fuel-soaked well is still a fuel-soaked well. Find another source until a test exists.

Shock chlorination recipes circulate widely, and some of them are in public pamphlets for shallow emergencies. They are easy to do wrong — too little and you grow confidence, too much and you have a hazardous waste in a hole, poured down a well that also connects to someone else's water. This edition will not print a "buckets of bleach per foot of casing" recipe. If a current local health sheet in your hand gives one, and you are the sort of well it applies to, that sheet outranks this book. If you are guessing, you are not disinfecting. You are seasoning.

Lightning is an electrical trade

A metal roof, a full tank, a wet soil, and a storm are a serious neighbourhood. The long document in the research pile is a design project against a national lightning code, full of down-conductors, earth electrodes, and separation distances. That is the point of mentioning it. Protection that works is designed and bonded by people who do that work, to a code that is still a code when the lights are out. A cable you ran from the gutter to the cistern "so the lightning has somewhere to go" can be the path through the tap, the stand, and the person holding the cup.

In a storm, stay off the roof, stay out of the bath if the house is the tallest thing in a field and the storm is on you, and do not become the tallest wet object out of curiosity. Tanks can be bonded into a real system. They should not be bonded into a theory you had during the thunder.

Backflow testing, septic repair after a flood, and the question of whether a town main is poisoned are the same shape of problem. They have professions. A field manual can tell you to wait, to use the sealed water, to boil what boiling can save, and not to invent a joint between a fire main and a kitchen. It cannot sign the certificate.

Wave at the jobs that are not yours. Then go and put the lid back on the barrel, which is.

CHAPTER 17

A Morning with Water



Plate 17. Dawn, a covered cup, and the boil still due.

The morning does not need a speech. It needs a cup you can explain to a person who was not there when you treated it.

If the tap is running and the authorities are happy, drink the tap, and use the rest of this book as practice and as a rain barrel's manners. If the tap is a rumour, walk the list without improvising.

Sealed water first, for drinking and for anyone small. If that is gone, the water you settled, then boiled for a minute — three if you are high — or dosed with unscented bleach whose percent you can read, then waited out. Roof water only after the first wash has been thrown away, and still treated. Fog water only as the small extra it is, and still treated. Nothing from the flood, nothing from the sprinkler, nothing that smells of a chemical, nothing from the fuel-can-that-is-now-a-jug because it is not.

The container is opaque, or in the dark, food-grade, dated, and closed. The ladle is clean. The note on the barrel says the day and the method in words a stranger can use. If you are pouring for a baby, you already chose boil-and-cool or a sealed bottle, and you did not get creative with iodine.

Gravity can bring it down the hill if the hill is high enough and the stand is a stand. The olla can keep the tomatoes from becoming a drama, and nobody drinks from the olla. The kettle can be hot. The coil on the stove is not your project this year.

What you tell the next person

Where it came from. What you did. What you did not do. When. Where the bleach is, and what percent the label claims, and that the label wins. Where the fuel for the kettle is, and that the burner lives outdoors or under a real flue. Who must not be given the iodine. Which tap is still "do not drink" because it talks to a well you have not tested.

Then you pour the cup, you put the lid back, and you do not leave the story in your head. Heads are where stories go to become rumours. Paper is cheaper than a second illness.

The dead may walk. The water still needs a boil. The roof still needs its first wash thrown away. The sprinkler pipe still is not a well. None of that is heroic. It is the reason the morning has a drink in it, and the reason the drink does not become the emergency.

Check the lid. Then quiet.

Afterword

The last chapter of a water book is a temptation to get solemn about civilisation. Resist it a little. The yard is not a chapel. It is a patch of ground you share with whoever needs a drink, a pot washed, or a garden persuaded through a dry week, and the courtesy that keeps it usable is older than any of us: know where it came from, treat it before you trust it, and leave a way to tell the next person that you did.

If you have read this far you already know the unglamorous truth. Clear is not the same as safe. Boiling is a kindness to germs' enemies and a uselessness against solvents. The question here is not how rustic the barrel can look. It is whether the next person can pour a cup, explain what was done to it, and put the lid back without becoming the incident. The person who brings unscented bleach with the percent still readable will be remembered more kindly than the person who brings a speech.

The zombies, if you still need them, are only the name we gave the interruption. A skipped boil is one. A fuel can reused as a jug is one. Panic that drinks the first wash off the roof and walks away is one. You do not beat them by adding gadgets. You beat them by being able to say, on paper, what the cup is supposed to be when the morning comes back.

Leave a note on the barrel. Tell someone else the date, the method, and where the lid lives. When you are done for the night, say so, and cover it on purpose, so nobody sits up listening to a drip out of loyalty.

The signature under an afterword belongs to the person who stands by the book. That line is blank until you want it.

Boil it. Then quiet.

AUTHOR

PLACE AND DATE

Acknowledgments

The author wishes to thank

NAME _____

NAME _____

NAME _____

and the person who said, once more, boil it before you blame the horde.

About the author

NAME

was born in

PLACE

Two or three sentences of biography. Leave this until you are ready to sign the book.

A LINE

ANOTHER

THEY LIVE IN

Also forthcoming

from

PUBLISHER

Placeholder titles, so the page has the right furniture. Replace them with the real list.

Zombieproof Radio

Volume I, already on the shelf. The air, if the bench ever lets you stand up. Thirty-eight chapters, and a flower in the antenna.

Zombieproof Computers

Volume II, already on the shelf. A mind that still remembers the list in the morning. Seventeen chapters, and a daisy in the drive.

Zombieproof Engines

Volume III, already on the shelf. How to keep the wheels turning, and the exhaust pointed at nobody you like. Seventeen chapters, and a daisy in the filler.

Colophon

Zombieproof Water: How to Keep a Cup When the Tap Goes Quiet was set in Fraunces and Literata. The plates were painted for this edition. The text was prepared from a pile of public-health notes, rainwater manuals, and a foundation draft on off-grid water, then dressed in the voice of the yard. The doses were kept where the notes gave them, and labelled as indicative. Nothing in the arithmetic was promoted to a universal recipe.

Trim: six by nine inches. The author, the dedication, and the publisher are intentionally blank.

If you are holding a file and not a book, print it on paper that can stand a damp bench. Leave the last page near the kettle, not in a drawer. The boil is still due.

PUBLISHER

CITY